Horizon

OxHack '26 · https://github.com/glebbabichevv/oxhack26

Polymarket prices are miscalibrated as a function of *time to resolution*. Does correcting that survive spreads, fees and an honest out-of-time test, and *when* is it worth acting?

1. Data

- 34,677 resolved **binary** Polymarket markets (Gamma + CLOB hourly price history). Cancelled / 50–50 resolutions dropped.
- One row per (market, horizon) with $h \in \{720, 168, 72, 24, 12\}$ hours before the **scheduled** `endDate` (live markets only know that; `closedTime` would leak early closes).
- Entry price = last trade print at or before the target time, prior prints only, ≤ 6 h old (98.9% ≤ 1 h). Prices kept in $[3\%, 97\%]$.
- Domain from event tags: sports, politics, crypto, culture, economics, science/tech, geopolitics, other. Lifetime volume $\ge$ \$100k.

2. Out-of-time split

- Train: events that ended before **2026-05-18** (23,838 markets). Test: the 10,579 after.
- Legs of one event never straddle the split; 260 legs that resolved before the cutoff dropped from test.
- Embargo: a test trade counts only if entered on/after the cutoff.
- Validation = last 20% of train events (model and rule selection happen here only).

3. The model (M0): 4 coefficients

With price p and hours h, let $\ell = \operatorname{logit} p = \ln \frac{p}{1-p}$ and $\tau = \ln \frac{\operatorname{clip}(h, 12, 720)}{24}$:

$$q \;=\; \sigma(\beta_0 + \beta_1\,\ell + \beta_2\,\tau + \beta_3\,\ell\,\tau)\,, \qquad \sigma(x) = \frac{1}{1+e^{-x}}$$

Fitted (logistic regression, train only): $\beta_0 = 0.066$, $\beta_1 = 0.845$, $\beta_2 = -0.015$, $\beta_3 = 0.022$.
Slope $\beta_1 + \beta_3\tau < 1$ near the end $\Rightarrow$ prices are **overconfident** there: longshots win more than priced, favourites less; a month out ($\tau \approx 3.4$) the slope is ≈ 0.92, about fair.
Uncertainty band. 40 refits on bootstrap resamples *by event*; $[q_{10}, q_{90}]$ = 10th/90th percentile of their predictions.

4. Costs

$$c = \max\!\left(\tfrac{\text{ask}-\text{bid}}{2},\ 1¢\right) + r\,\big(p(1-p)\big)^{e}$$

Polymarket taker fee with each market's own `feeSchedule` (rate r, exponent e; default rates 4–7% by domain). Historical spreads are unavailable, so the 1¢ floor is a stated limitation.

5. Decision rule

Net edge on the conservative end of the band:

$$e_{\text{YES}} = q_{10} - (p + c), \qquad e_{\text{NO}} = (p - c) - q_{90}$$

Back YES if $e_{\text{YES}} > 0$, NO if $e_{\text{NO}} > 0$, else no trade. **Policy V3** adds: net edge ≥ 1¢, and only within **24 h** of the end (earlier $\Rightarrow$ WAIT). No call under 12 h (never trained or tested there).
Stake: quarter Kelly, capped. For YES at cost $\pi = p + c$:

$$f = \min\!\left(\tfrac{1}{4} \cdot \frac{q_{10} - \pi}{1 - \pi},\ 5\%\right)$$

Open stake ≤ 1 starting bankroll (later entries get what is left, pro rata at ties). P&L additive, no compounding.

6. Policy COH (shipped)

In a neg-risk event whose legs at one horizon number ≥ 2 and whose prices sum to $[0.95, 1.25]$:

$$q_i \leftarrow \frac{q_i}{\sum_j q_j}, \qquad \text{band offsets } q_i - q_{10,i},\ q_{90,i} - q_i \text{ kept}$$

then V3, then **one bet per event** (largest net edge; ties broken by market id, stable sort). Thin legs (< \$100k) price the book but get no call.

7. Metrics

$$\text{Brier} = \tfrac{1}{n}\textstyle\sum (q - y)^2$$

$$\text{LogLoss} = -\tfrac{1}{n}\textstyle\sum \big[y \ln q + (1-y)\ln(1-q)\big]$$

Return per \$1 staked $= \sum_k f_k r_k \big/ \sum_k f_k$, with payoff $r = \frac{y}{p+c} - 1$ (YES) or $\frac{1-y}{1-p+c} - 1$ (NO). 90% intervals: bootstrap resampling *events* (legs of a match move together), 1,000–2,000 draws, seed 0.

8. Results (test, out of time)

	per \$1 [90% CI]	trades	max DD
COH	**+30.5¢** [21.7, 39.5]	1,827	0.22
V3	+17.2¢ [11.8, 23.5]	3,653	0.38
V0 (no rule)	+11.1¢		
Back the favourite	−8.70 bankrolls		

Brier 0.1297 (model) vs 0.1305 (market); log loss 0.3917 vs 0.3940. COH: +4.27 bankrolls [3.05, 5.53], 30.7% of bets won.
Where: sports +33¢ (1,241 bets), politics +75¢ (103), culture −14¢ (the one losing domain), crypto ≈ 0.
When (mean return per trade, V3): 7d +7.8%, 3d +8.8%, 1d +13.8%, 12h +18.5%; 30d no trade clears 1¢.
The cited paper (arXiv 2602.19520) found prices compressed toward 50% far out; **we did not reproduce that direction**.

9. Everything we tried (selected on validation only)

Candidate	Outcome
M0 (above)	shipped model
M1 logistic + momentum, legs, book share	no gain over M0
M2 market-anchored boosting	no gain over M0
Beta calibration (Kull 2017)	$\le$ M0
Isotonic per horizon	−7.8¢ vs M0 (walk-forward)
Hierarchical domain model	domain terms overfit
Shin (1993) de-vig	−10¢: gets Polymarket backwards
Power method	$\approx$ market
Mean ensemble / log pool	errors correlate 0.99, no gain
Agreement filter	fewer trades, no gain
Rules V0–V3	V3 picked (+25.5¢ val)
COH	**picked** (paired +8.8¢ [0.7, 17.3] val)
Dynamic exit (sell early)	no extra profit; DD 0.20 vs 0.23
Crypto as options (Binance)	Brier 0.1471 vs 0.1399, −16.5¢

10 variants on validation; test opened once for V3 and COH side by side (exit: a second, descriptive look).

10. Robustness

- Stale prices: only prints ≤ 1 h old and markets closed ≥ 12 h after the print: COH +29.6¢ [21.1, 39.7].
- Placebo "prices are right" ($y \sim$ Bernoulli(p), 1,000 draws, same trades): −8.1¢ median vs real +31¢, $p < 0.001$.
- Walk-forward by month (refit on earlier events): COH positive in 15 of 17 months.
- Deflated Sharpe 0.9999 over 12 trials (per-trade SR 0.135).

- Survives double Polymarket fees (+15.7¢ V3). Capacity ≈ $10k per market on liquid football books.
- The gain is one bet per event, not leg picking: a random eligible leg gives +33.5¢.

11. Sealed predictions

Live predictions frozen to a file; its SHA-256 posted in the event Discord before the markets closed and anchored in Bitcoin via OpenTimestamps (Seal 1 block 972944, Seal 2 block 972999). Pre-registered metric: Brier vs the market with an event-bootstrap 90% interval; only markets resolved after the post count.

12. Reproduce

`python integrations/reproduce.py` — re-runs the pipeline, 13/13 headline numbers.

`python integrations/backtest_log.py` — every test trade as CSV.

`python integrations/verify_seals.py` — hashes + Bitcoin blocks.

`node extension-module/parity.mjs <seal>` — extension maths = Python.